

Quantum-Chemistry Inspired Approach to Ultra-Cold Atoms in Tweezer Arrays
(Multi-Species Generalisation of SCF, FCI, CASCI)

Lyding Brumm @ Humboldt-Universitaet zu Berlin

About Me

- B. Eng. in Telecommunication Engineering (2019)

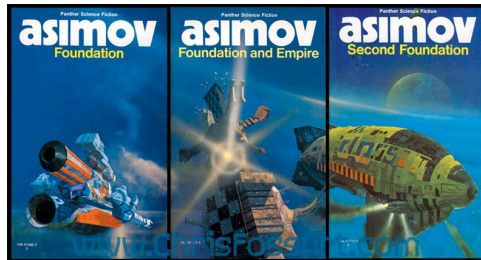
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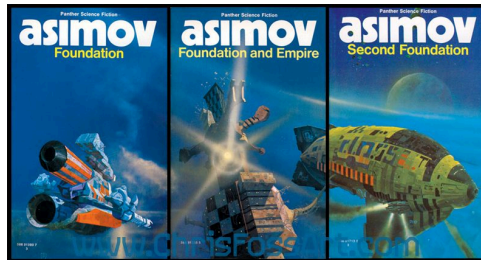
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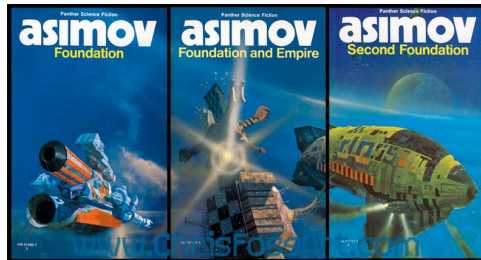
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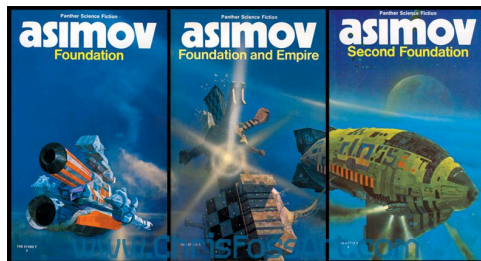
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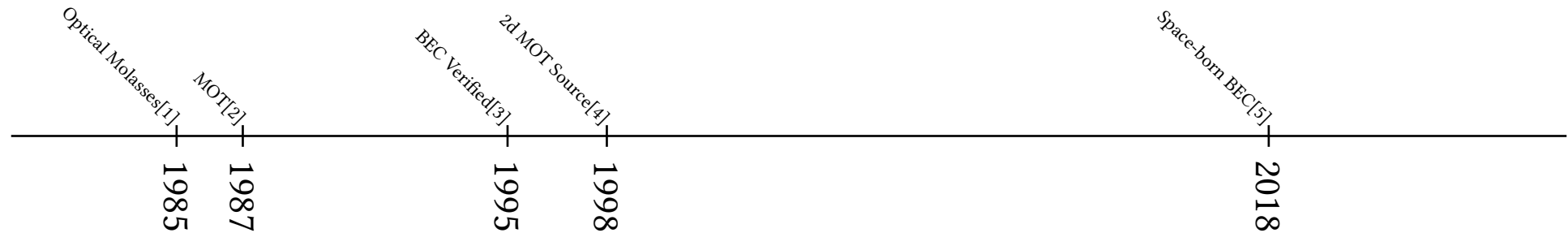


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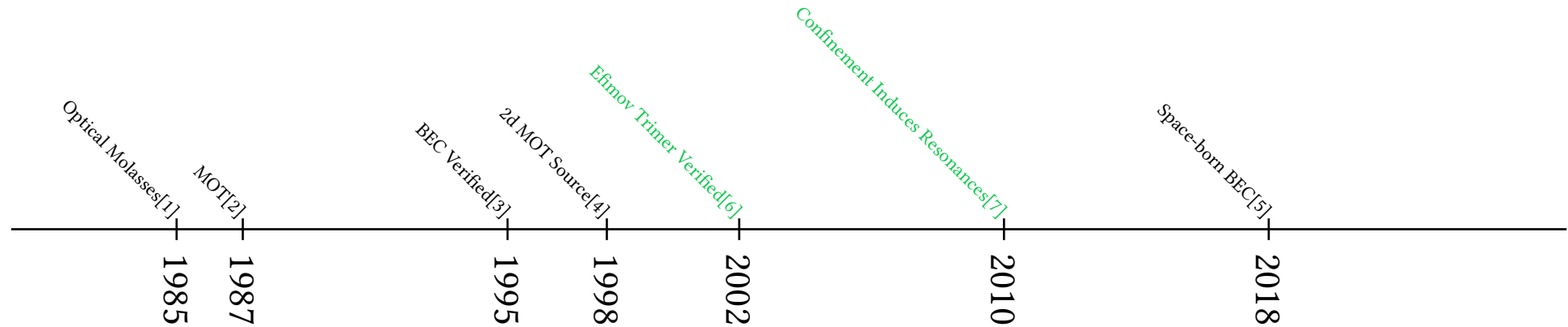
History and Motivation of Ultra-Cold Atoms

black: General, red: Quantum Simulator, blue: Quantum Computer, green: Few Body Physics, purple: Numerics



- [1] S. Chu *et al.*, “Three-dimensional viscous confinement and cooling of atoms by resonance radiation pressure,” *Phys. Rev. Lett.*, 1985.
- [2] E. L. e. a. Raab, “Trapping of Neutral Sodium Atoms with Radiation Pressure,” *Phys. Rev. Lett.*, 1987.
- [3] M. H. A. el al., “Observation of Bose-Einstein condensation in a dilute atomic vapor,” *Science*, 1995.
- [4] K. Dieckmann *et al.*, “Two-dimensional magneto-optical trap as a source of slow atoms,” *Phys. Rev. A*, 1998.
- [5] D. Becker *et al.*, “Space-borne Bose–Einstein condensation for precision interferometry,” *Nature*, p. 391, 2018.

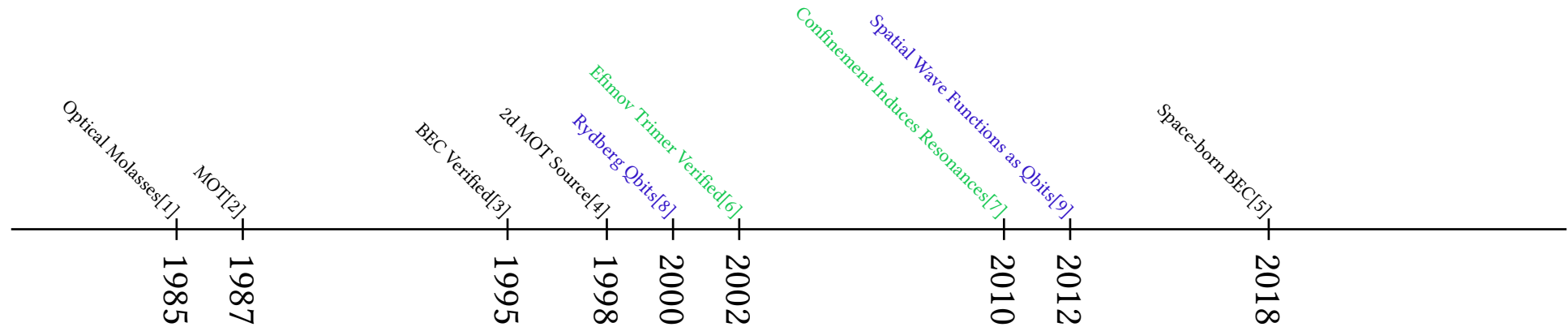
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[6] T. Kraemer *et al.*, “Evidence for Efimov quantum states in an ultracold gas of caesium atoms,” *Nature*, vol. 440, p. 315, 2006.

[7] H.-C. Nägerl *et al.*, “Confinement-Induced Resonances in Low-Dimensional Quantum Systems,” *Phys. Rev. Lett.*, 2010.

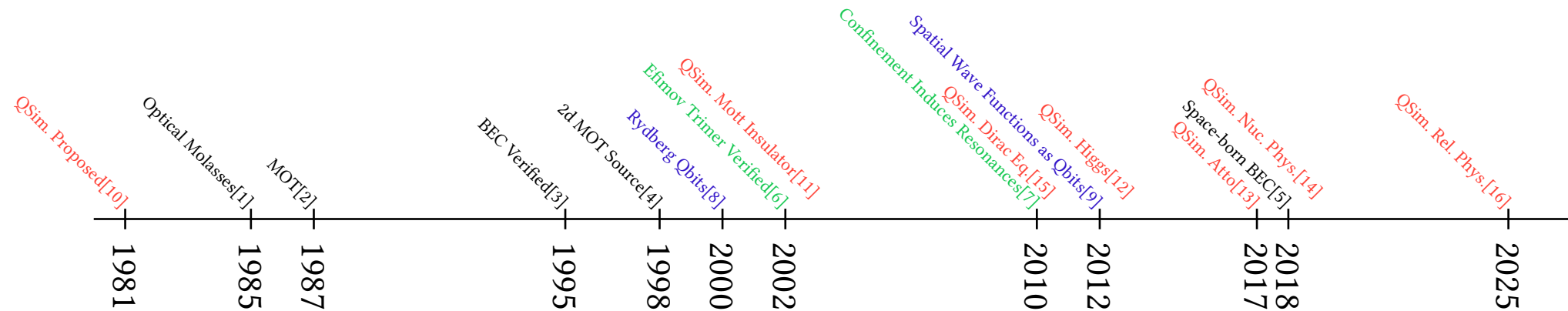
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[8] M. Lukin *et al.*, “Fast Quantum Gates for Neutral Atoms,” *Physical review letters*, 2000.

[9] P.-I. Schneider *et al.*, “Quantum computation with ultracold atoms in a driven optical lattice,” *Phys. Rev. A*, 2012.

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[10] F. R., “Simulating physics with computers,” *International Journal of Theoretical Physics*, 1982.

[11] I. Bloch *et al.*, “Quantum phase transition from a superfluid to a Mott insulator in a gas of ultracold atoms,” *Nature*, 2002.

[12] I. Bloch *et al.*, “The ‘Higgs’ amplitude mode at the two-dimensional superfluid/Mott insulator transition,” *Nature*, vol. 487, 2012.

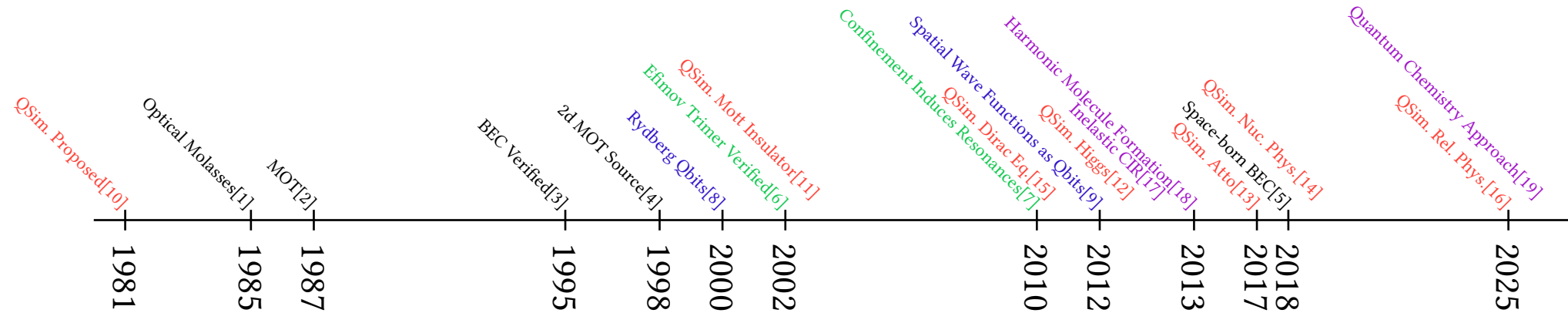
[13] S. Sala *et al.*, “Ultracold-atom quantum simulator for attosecond science.” p. 11403, 2017.

[14] E. Rico *et al.*, “SO(3) “Nuclear Physics” with Ultracold Gases,” *Annals of Physics*, vol. 393, pp. 466–483, 2018.

[15] C. F. Roos *et al.*, “Quantum simulation of the Dirac equation,” *Nature*, 2010.

[16] R. Schützhold, “Ultra-cold atoms as quantum simulators for relativistic phenomena,” *Progress in Particle and Nuclear Physics*, vol. 145, 2018.

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[17] S. Sala *et al.*, “Inelastic Confinement-Induced Resonances in Low-Dimensional Quantum Systems,” *Physical Review Letter*, 2012.

[18] S. Sala *et al.*, “Coherent molecule formation in anharmonic potentials near confinement-induced resonances,” *Physical Review Letter*, 2013.

[19] M. u. Rehman *et al.*, “Confined few-particle systems beyond mean-field theory adopting Gaussian-type orbitals and Morse interparticle interaction,” *Physical Review A*, vol. 112, p. 63320, 2025.

Quantum Simulator

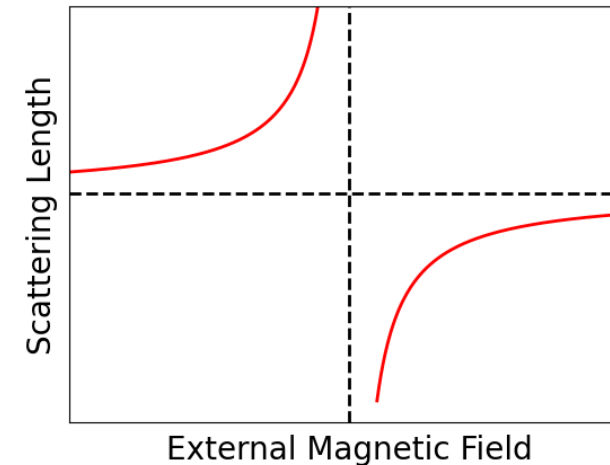
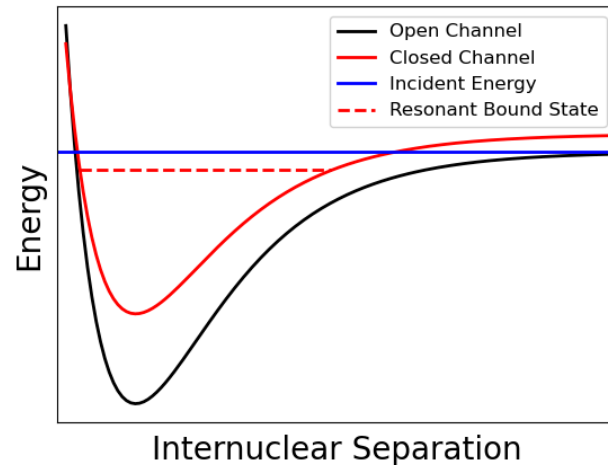
- Optical Lattices, Orthogonal (reflected) Laser Beams

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- Optical Tweezer Arrays, Highly focused Laser Beam

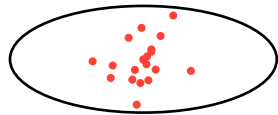
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- Feshbach Resonances $\rightarrow$ Tunable (effective) interaction-strength

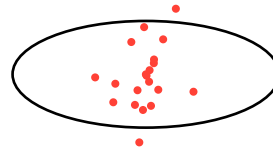
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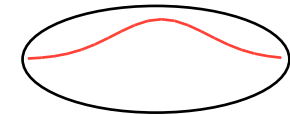
$$H = -J \sum_{\langle i,j \rangle} a_i^\dagger a_j + \sum_i e_i n_i + \frac{1}{2} U \sum_i n_i (n_i - 1)$$



Step 1: Load atoms into cigar shaped trap

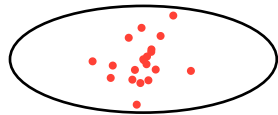


Step 2: Evaporative cooling

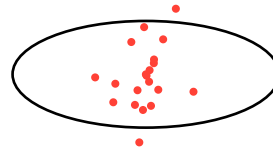


Step 3: BEC Formation

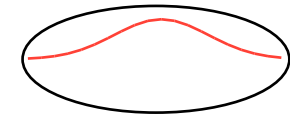
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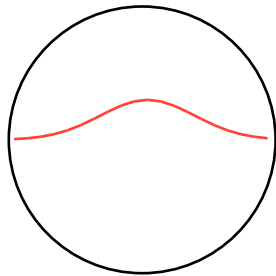
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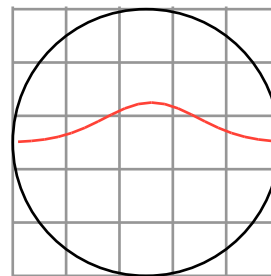
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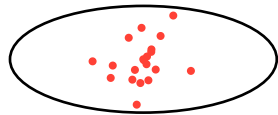


Step 4: Relax trap to isotropic trap

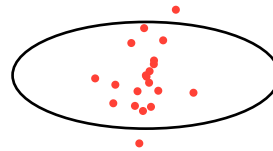


Step 5: Add optical lattice

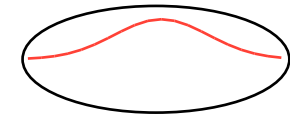
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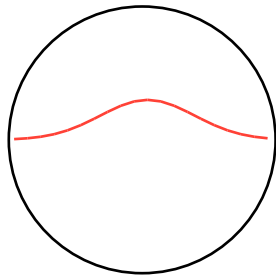
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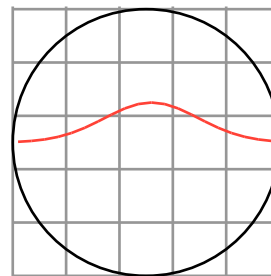
Step 2: Evaporative cooling



Step 3: BEC Formation



Step 4: Relax trap to isotropic trap

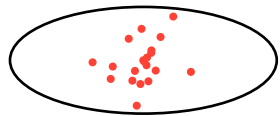


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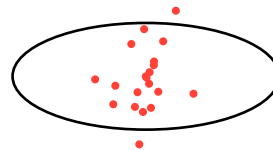


Step 6: Analyse interference pattern

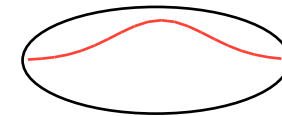
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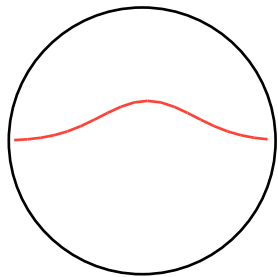
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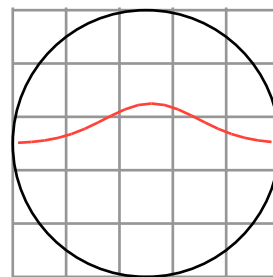
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Step 3: BEC Formation



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- Inter-atomic Potential

Quantum-Chemistry

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Ultra-Cold Atoms

- Few Body
- Non-relativistic => Schrödinger Equation
- Variable Geometry
- Inter-atomic Potential
- Bosons and Fermions

Quantum-Chemistry

- Few Body
- Non-relativistic => Schrödinger Equation
- Variable Geometry
- Coulomb Interaction
- Indistinguishable Fermions

$$\varphi_{\text{gto}}^{\sigma,R,\alpha} = N \prod_{j=1}^3 (x_j - R_j)^{\sigma_j} \exp\left(-\alpha(x_j - R_j)^2\right)$$

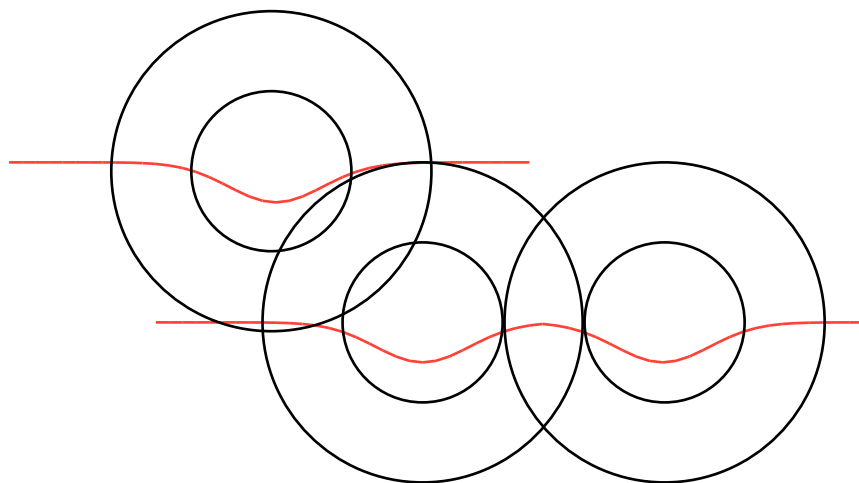
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Ultra-Cold Atoms (red: traps)

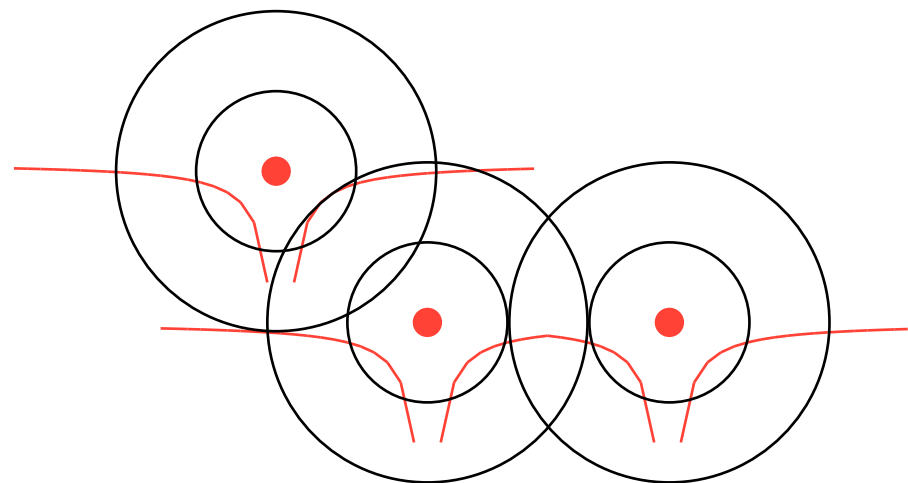
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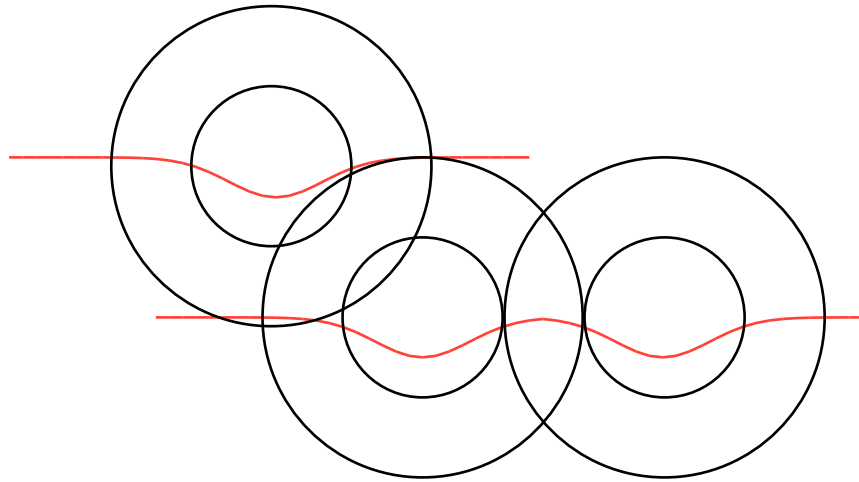


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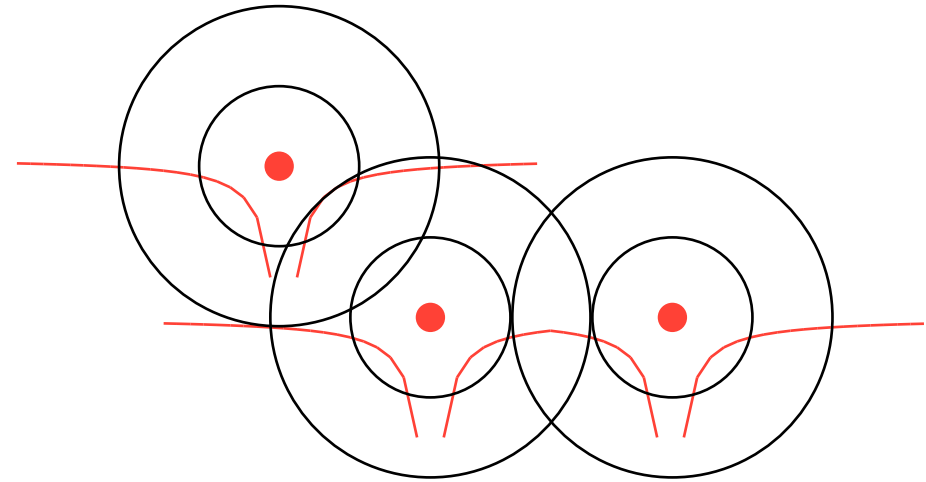
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Ultra-Cold Atoms (red: traps)



CGTOs are phys. motivated

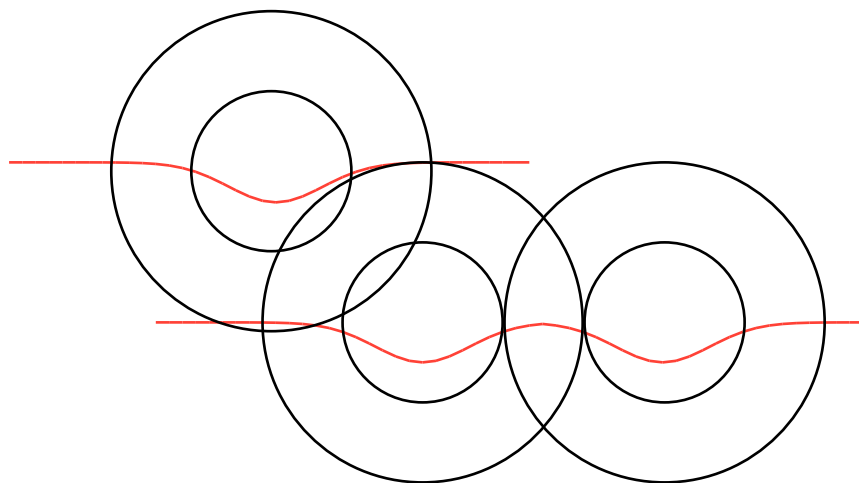
Quantum-Chemistry (red: cores)



CGTOs are **not** phys. motivated

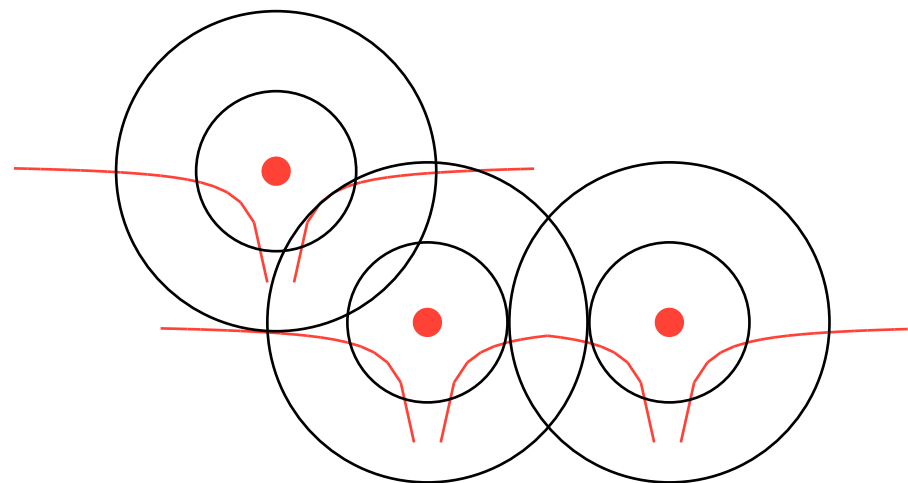
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Ultra-Cold Atoms (red: traps)



CGTOs are phys. motivated

Quantum-Chemistry (red: cores)



CGTOs are **not** phys. motivated

Traps $\leftrightarrow$ Cores
Atoms $\leftrightarrow$ Electrons

Where we started & What we focus on

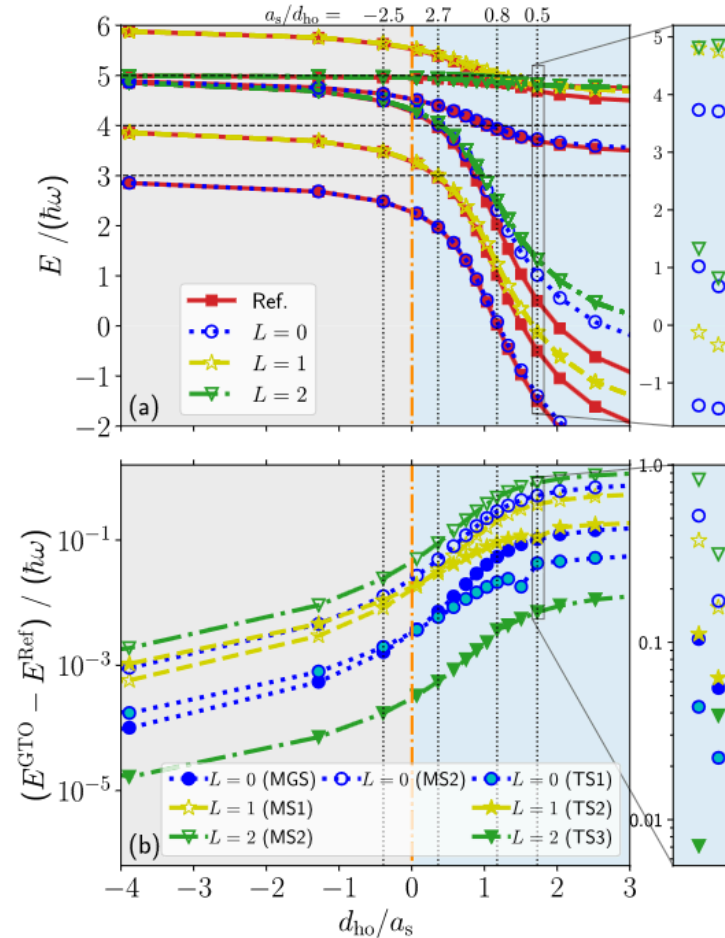
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- Spectrum as a function of the tweezer positions
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Morse:
 Depth: [0.5, 15]
 Minimum at: 0.3
 Dampening²: 10

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Multi-Species Generalisation (MSSCF, MSFCI)

- Use product wave ansatz

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- Minimize energy functional w.r.t single-particle wave function
- Coupled Fock equation for every species
- Extended Pople-Nesbet equation $\Rightarrow$ Internal consistency check of the theory

$$H = \sum_i^{\mathcal{N}_B} \sum_{n=1}^{N_i} h_i(r_{i,n}) + \sum_i^{\mathcal{N}_F} \sum_{n=1}^{N_i} h_i(r_{i,n})$$

$\mathcal{N}_B, \mathcal{N}_F$: number of bosonic and fermionic species N_i : number of particles in i th species

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& - \sum_i^{\mathcal{N}_B} \sum_j^{\mathcal{N}_F} \sum_{n,m}^{N_i, N_j} w^{(ij)}(r_{i,n}, r_{j,m})
\end{aligned}$$

$\mathcal{N}_B, \mathcal{N}_F$: number of bosonic and fermionic species N_i : number of particles in i th species

- Fermions: Well known Hartree-Fock ansatz

$$|\mathcal{F}^i\rangle = \sum_{\sigma \in \mathcal{S}_N} (-1)^{\# \sigma} \varphi_{\sigma(1)}(r_1) \cdots \varphi_{\sigma(N)}(r_N)$$

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- Bosons: All Bosons are in ground-state

$$|\mathcal{B}^i\rangle = \varphi_1(r_1) \varphi_1(r_2) \dots \varphi_1(r_N)$$

Bosonic Species

Fermionic Species

Bosonic Species

$$\begin{aligned}\mathbb{F}_{\text{bos}}^{\kappa}(c^1, \dots, c^{\mathcal{N}}) &= \mathbb{h}_{km}^{\kappa} + (N_{\kappa} - 1) \sum_{l,n} \mathbb{w}_{klmn}^{\kappa\kappa} c_l^{\kappa} c_n^{\kappa} \\ &+ \sum_{i \neq \kappa}^{\mathcal{N}_B} N_i \sum_{l,n} \mathbb{w}_{klmn}^{i\kappa} c_l^i c_n^i \\ &+ \sum_{i=\mathcal{N}_B+1}^{\mathcal{N}} \sum_{z=1}^{N_j} \sum_{l,n} \mathbb{w}_{klmn}^{j\kappa} c_l^{j,z} c_n^{j,z}\end{aligned}$$

Fermionic Species

$$\begin{aligned}\mathbb{F}_{\text{fer}}^{\kappa}(c^1, \dots, c^{\mathcal{N}}) &= \mathbb{h}_{km}^{\kappa} + \sum_{z=1}^{N_{\kappa}} \sum_{l,n} (\mathbb{w}_{klmn}^{\kappa\kappa} - \mathbb{w}_{klnm}^{\kappa\kappa}) c_l^{\kappa,z} c_n^{\kappa,z} \\ &+ \sum_{i=\mathcal{N}_B+1}^{\mathcal{N}} \sum_{z=1}^{N_i} \sum_{l,n} \mathbb{w}_{klmn}^{i\kappa} c_l^{i,z} c_n^{i,z} \\ &+ \sum_{i=1}^{\mathcal{N}_b} N_i \sum_{l,n} \mathbb{w}_{klmn}^{i\kappa} c_l^i c_n^i\end{aligned}$$

Bosonic Species

Fermionic Species

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Non-trivial point group $\mathcal{G}$

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$$|\mathcal{B}^i\rangle = \sum_{\sigma \in (\mathcal{S}_n/\mathcal{G})} \sum_{\rho \in \mathcal{G}} \varphi((\sigma.\rho)(1)) \dots \varphi((\sigma.\rho)(N_i)), \quad \mathfrak{B} = \prod_i^{\mathcal{N}} \mathfrak{B}^i, \quad \text{Fermions} \subset \text{Bosons}$$

Class	Differences
Diag.	The wave functions are identical
BFON	At one index in a bos. species
FFON	At one index in a fer. species
BSON	At two indices in the same bos. species
FSON	At two indices in the same fer. species
BBSON	At two indices in different bos. species
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case	H_1	H_2	H_3	H_4	H_5	H_6	H_7
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BFON	X		X		X		X
FFON		X		X		X	X
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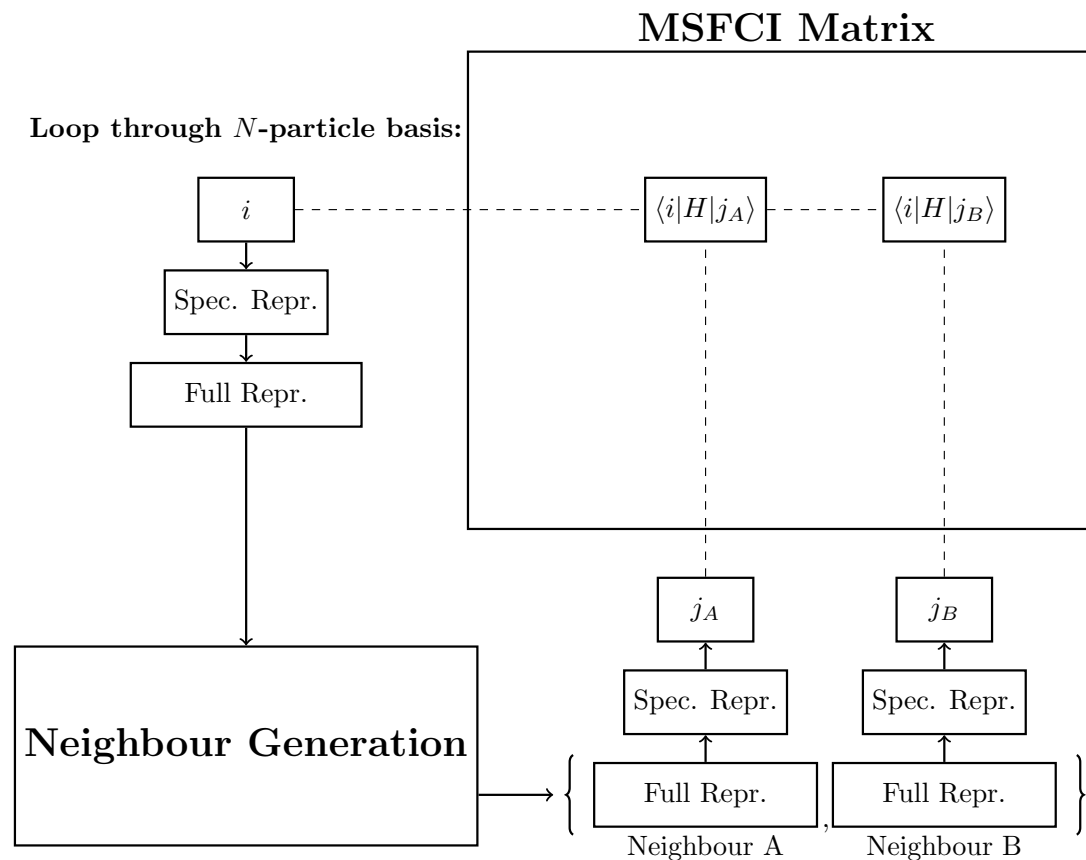
20 cases of brackets which have to be derived

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Verification and Preliminary Results

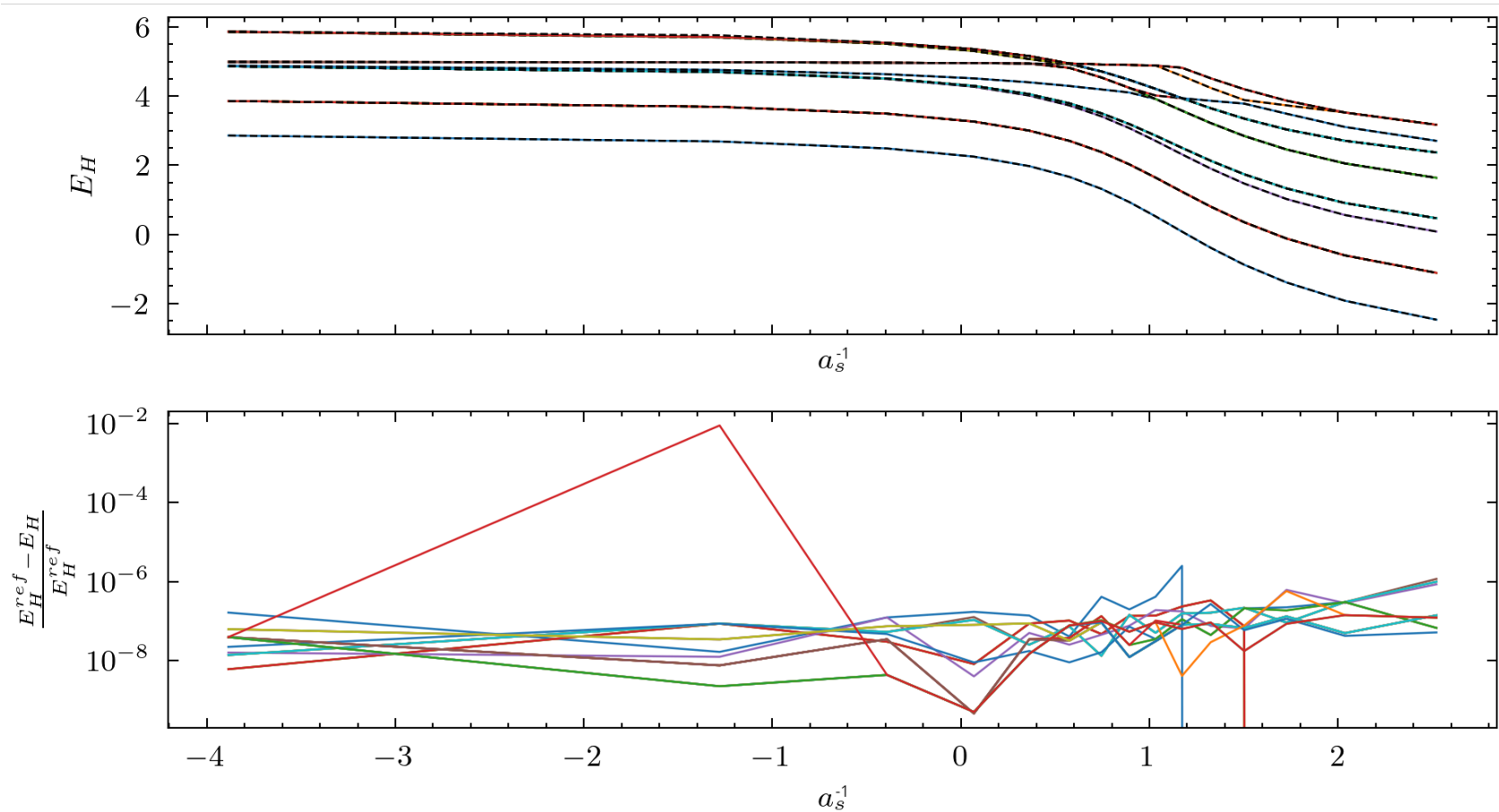


Figure 1: 2 Boson data (from [19] M. u. Rehman *et al.*, “Confined few-particle systems beyond mean-field theory adopting Gaussian-type orbitals and Morse interparticle interaction,” *Physical Review A*, vol. 112, p. 63320, 2025.) reproduced. **Solid: Literature data. Dashed: MSFCI data**

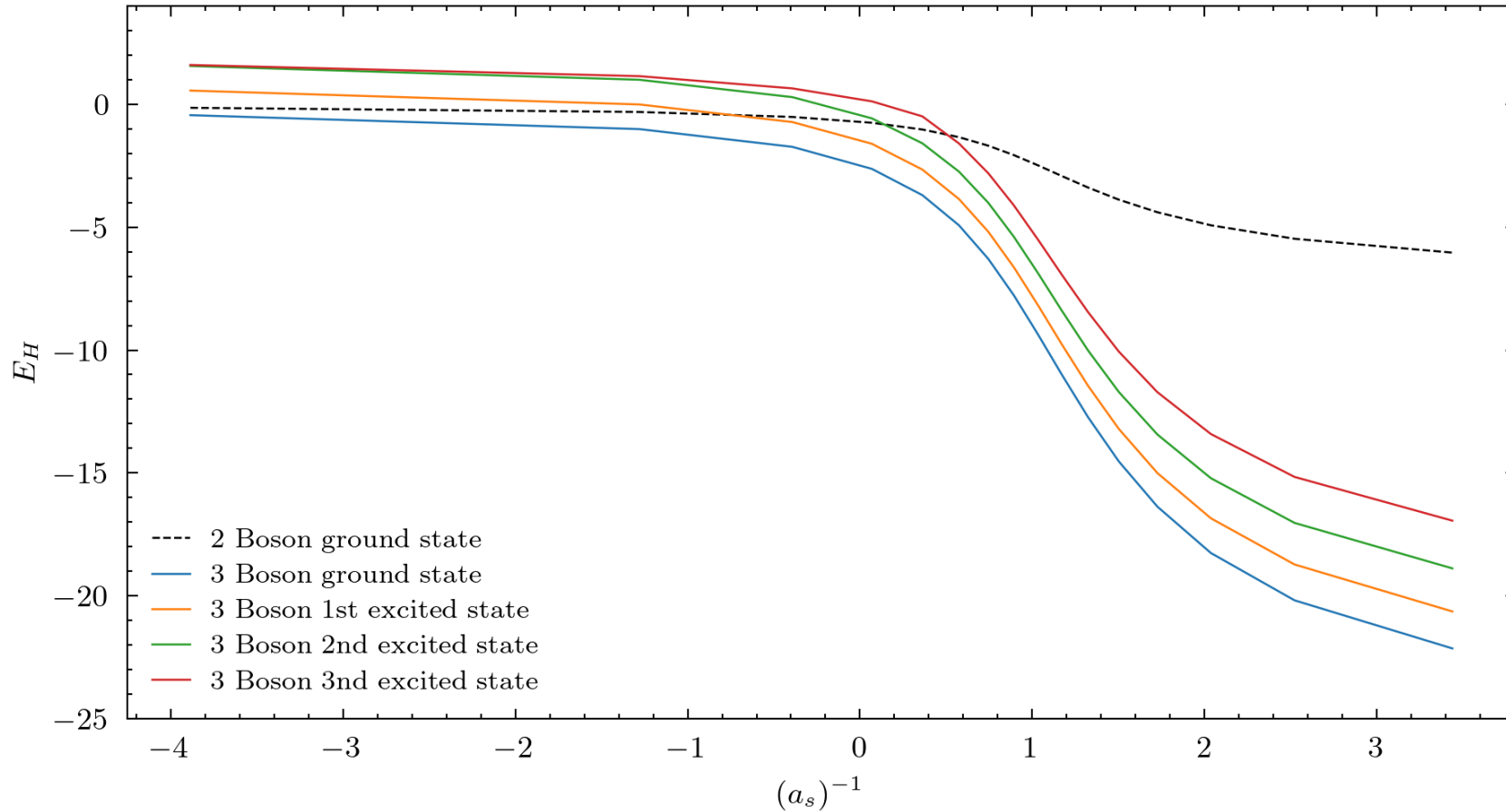


Figure 2: 2 Boson / 3 Boson Comparison

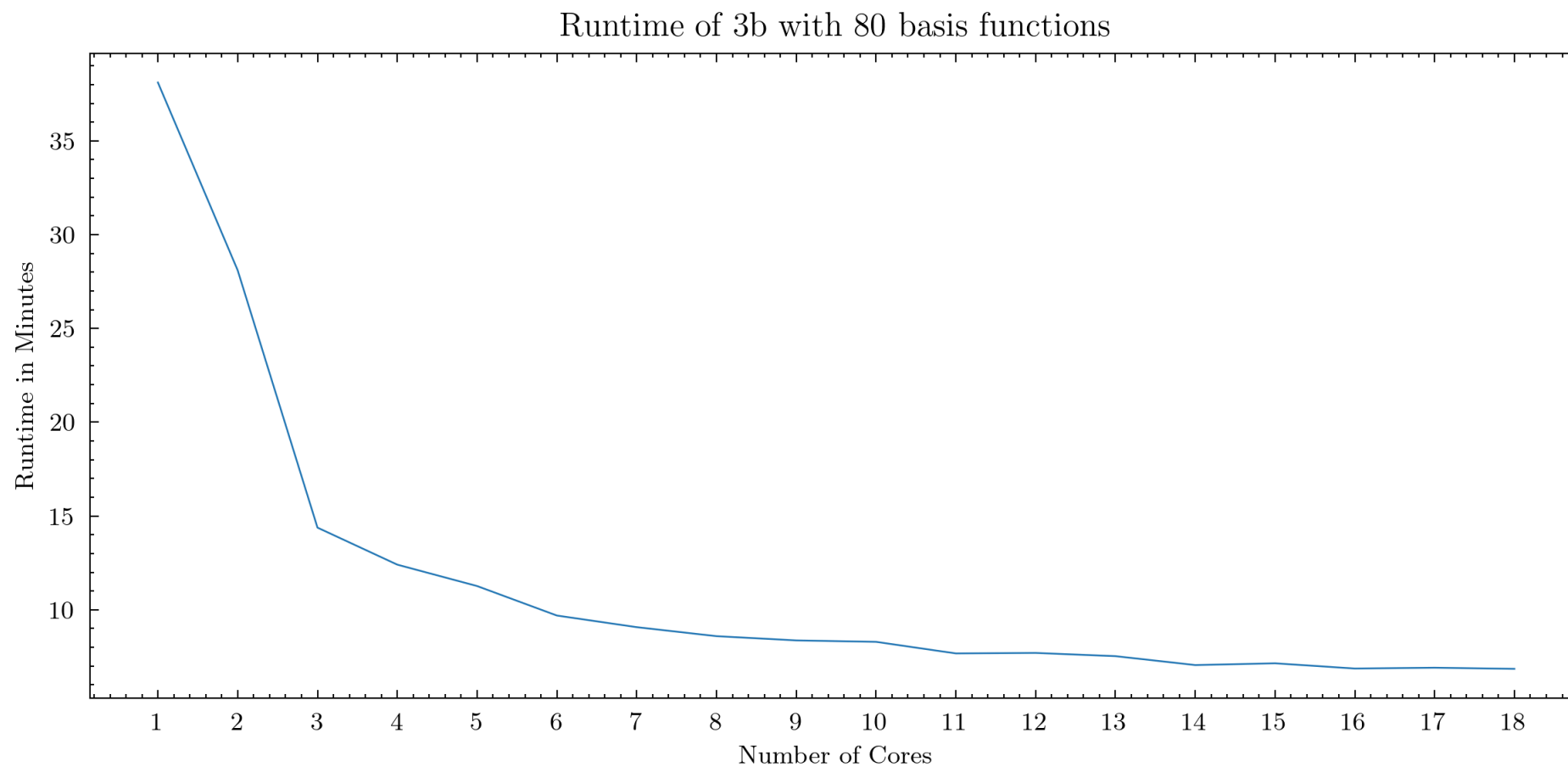


Figure 3: 2 Boson / 3 Boson Comparison

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